A RESEARCH PROJECT ON FINANCIAL INFORMATION RETRIEVAL

FinAlign.

Bridging the financial vocabulary gap between people and documents.

UNIVERSITÀ DI PAVIA
DI PILATO · CINQUEMANI · MARCHIONI
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THE PROBLEM

People and documents *don't speak the same language.*

Traditional search relies on literal keyword matching — and fails the non-experts who need financial information most.

USER ASKS	"Should I put money in property funds?"
DOC SAYS	"REIT allocation under current yield curve assumptions..."
USER ASKS	"Is this a good time to buy a house?"
DOC SAYS	"30-year fixed mortgage spreads vs. treasuries..."

"Does query rewriting, prefix-based encoding, and document chunking improve the effectiveness of a hybrid `BM25 + dense` *retrieval system for financial questions?"*

THE DATASET

FiQA, from the *BEIR benchmark.*

57,600	648	1,705
DOCUMENTS	QUERIES	QRELS
News headlines, microblog posts, and report snippets.	Natural-language financial questions from non-experts.	Graded relevance annotations for evaluation.

"What is the difference between a REIT and a real estate stock?"

EXAMPLE QUERY
CLEANED BEIR / FIQA TEST SPLIT

THE PIPELINE

From a question to *a ranked list.*

01

Query Rewriting *& HyDE*

An LLM expands the user query with technical financial vocabulary and drafts a hypothetical answer.

`user query` → `original + rewrites + HyDE draft`

02

Hybrid *BM25 + BGE*

BM25 retrieves on the lexical index. BGE retrieves on a chunked dense index, with chunk scores max-pooled to the document level. Both runs are fused (RRF or linear).

`enriched queries` → `fused top-K candidates`

03

Document *Chunking*

Documents are split into overlapping passages before dense indexing — exposing localized relevance signals to the bi-encoder.

`long document` → `passages → max-pool → doc score`

04

MonoT5 *Re-ranking*

A cross-encoder rescores the top candidates by jointly encoding query–document pairs.

`top-K candidates` → `final ranking`

DATA FLOW

Q · "Is buying a house a good investment?"

↓

01 rewrite + HyDE
→ original · rewrite · hypothetical

↓

02a BM25
lexical, full docs

02b BGE dense
on chunks → max-pool

↓ fuse ↓

03 top-K candidates

↓

04 MonoT5 re-rank

↓

R · ranked documents

END-TO-END RETRIEVAL PIPELINE

QUERY REWRITING & HYDE

One question, *many semantic views.*

An LLM rewrites the query with technical financial vocabulary, then drafts a *hypothetical answer* (HyDE) used as a semantic proxy.

Original, rewrites, and HyDE are fused via score combination — diversification, not substitution.

ORIGINAL	"Is buying a house a good investment right now?"
REWRITE	"Residential real estate purchase ROI under current mortgage rates and housing market conditions."
HYDE	"Buying a house can be a good investment depending on local price-to-rent ratios, mortgage spreads, and expected holding period…"

HYBRID RETRIEVAL

Two retrievers, *one ranking.*

SPARSE

BM25

First-stage lexical retrieval. Term saturation, document-length normalization, exact-match recall.

Index · Terrier inverted
Strength · Rare terms, exact matches

DENSE

BGE bi-encoder

Semantic retrieval over BGE embeddings. Captures paraphrases and implicit financial concepts beyond lexical overlap.

Index · FAISS vector store
Strength · Synonyms, intent, paraphrase

QUERY PREFIX "Represent this sentence for searching relevant financial passages:"

↓ WHAT WE FOUND In our experiments, fusing BM25 with instruction-tuned BGE **did not improve over BGE alone**. The dense retriever already captured most of the signal — see slide 10.

DOCUMENT CHUNKING

Retrieve passages, *rank documents.*

Long documents dilute embeddings. Splitting into overlapping fixed-length passages exposes localized relevance signals to the dense retriever.

Chunk scores are aggregated to the document level using `max pooling` — one strong passage promotes its parent document.

DOCUMENT · SPLIT INTO CHUNKS

Introductory paragraph on portfolio diversification… CHUNK 01

Direct discussion of REIT vs. real-estate stock structure, dividend treatment, and liquidity profile… CHUNK 02

Tangential commentary on tax filing dates… CHUNK 03

Closing remarks and disclaimer… CHUNK 04

↓ max-pool over chunks ↓

Document score $= \max(s_1, s_2, s_3, s_4)$

NEURAL RE-RANKING

A cross-encoder for *fine-grained relevance.*

MonoT5 jointly encodes each query–document pair, modeling interactions a bi-encoder cannot. Applied only to the top candidate set for tractability.

Late-stage relevance modeling captures nuanced financial reasoning — and amplifies the gains from chunking.

TOP-K BEFORE → AFTER MONOT5

#1	doc_4188 · "REIT vs. real estate stock — what to know"	▲ 4
#2	doc_0921 · "Diversification basics for retail investors"	▼ 1
#3	doc_7704 · "Dividend treatment of REIT distributions"	▲ 2
#4	doc_1156 · "How real estate IPOs are structured"	▼ 2
#5	doc_3329 · "Tax filing dates for K-1 issuers"	▼ 3

RESULTS

Each stage compounds the gain.

#	PIPELINE	MAP	NDCG@10	R@10	P@5
01	TF-IDF	0.208	0.247	0.298	0.121
02	BM25 (lexical baseline)	0.213	0.252	0.305	0.124
03	BGE dense	0.301	0.349	0.418	0.171
04	BGE + instruction prefix	0.327	0.378	0.446	0.183
05	Hybrid BM25 + BGE · RRF fusion ↓ DROP	0.265	0.308	0.371	0.151
06	Hybrid BM25 + BGE · linear fusion ↓ SLIGHT DROP	0.319	0.371	0.439	0.180
07	BGE + prefix + LLM rewriting & HyDE	0.345	0.402	0.481	0.194
08	+ Document chunking	0.371	0.428	0.512	0.207
09	+ MonoT5 re-ranking · winning configuration	0.412	0.471	0.547	0.226

FINDING Adding BM25 to a strong dense retriever **hurt performance** — RRF most of all. The wins came from **prefixing, rewriting, chunking, and MonoT5**, not from lexical fusion.

LESSONS

Three things worth remembering.

01

Read the model's *training paradigm.*

Prefixing queries with the right retrieval instruction unlocked a step-change in BGE performance.

02

Rewriting is *diversification,* not substitution.

Combining the original query with rewrites and HyDE drafts beat replacing it. Multiple semantic views cover ambiguous intent.

03

More signals isn't always *more signal.*

Adding BM25 to a strong dense retriever introduced noise. Hybrid wins came from chunking and re-ranking, not from fusion alone.

FUTURE WORK

From retrieval to *answers.*

→ 01 **Tune fusion weights systematically.**

→ 02 **Difficulty-aware re-ranking.**

→ 03 **Extend to retrieval-augmented generation.**

The next step is a system that doesn't just rank documents — it answers questions, in plain language, for non-experts.

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Clear access to financial information helps people make better life decisions.

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↗ READ THE PAPER ↗ CONNECT ON LINKEDIN ↗ GITHUB · PDMDP/FINALIGN

Thanks for reading.